Operations Research 1: Deterministic Models
Linear Programming: Sensitivity Analysis

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*Primal Simplex and Dual Simplex Methods

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$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

$$\begin{aligned} \max \quad & z \\ \text{s.t.} \quad & z - c^T x = 0 \\ & Ax = b \\ & x \geq 0. \end{aligned}$$

$-c^T$	0
A	b

$-c_B^T$	$-c_N^T$	0
B	N	b

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \geq c \\ & \pi \ \ u.r.s. \end{aligned}$$

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$$x_B \geq 0$$

- What is the optimality condition for the current solution?

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$$x_B \geq 0$$

- If $\bar{c}_N^T \leq 0$, the current solution is optimal.

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$$x_B \geq 0$$

- Else, assume $\bar{c}_j > 0$.

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$x_B \geq 0$

$\bar{c}_j > 0$

$-\bar{c}_j$	$c_B^T x_B$
u_1	$x_{B(1)}$
$\vdots$	$\vdots$
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$\vdots$	$\vdots$
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What if $u_i \leq 0$, for all i?

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I	$B^{-1} N$	x_B

$x_B \geq 0$

$\bar{c}_j > 0$

$$\lambda^* = \min_{\{i | u_i > 0\}} \frac{x_{B(i)}}{u_i}$$

$-\bar{c}_j$	$c_B^T x_B$
u_1	$x_{B(1)}$
$\vdots$	$\vdots$
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$$\pi^T = c_B^T B^{-1}$$

$$\begin{array}{cc|c} 0 & -\bar{c}_N^T & c_B^T x_B \\ \hline I & B^{-1} N & x_B \end{array}$$

$$\bar{c}_N \leq 0$$

$$x_B \geq 0$$

$$\bar{c}_j > 0$$

$$\begin{array}{c|c} -\bar{c}_j & c_B^T x_B \\ \hline u_1 & x_{B(1)} \\ \vdots & \vdots \\ u_l & x_{B(l)} \\ \vdots & \vdots \\ u_m & x_{B(m)} \end{array}$$

$$\begin{array}{c|c} 0 & c_B^T x_B + \lambda^* \bar{c}_j \\ \hline 0 & x_{B(1)} - \lambda^* u_1 \\ \vdots & \vdots \\ 1 & \lambda^* \\ \vdots & \vdots \\ 0 & x_{B(m)} - \lambda^* u_m \end{array}$$

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- If $x_B \geq 0$, the current solution is both primal and dual optimal.

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- Else, assume $x_{B(l)} < 0$.

$\bar{c}_j > 0$

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$-\bar{c}_1$	$\dots$	$-\bar{c}_j$	$\dots$	$-\bar{c}_n$	$c_B^T x_B$
v_1	$\dots$	v_j	$\dots$	v_n	$x_{B(l)}$

$x_{B(l)} < 0$

What if $v_i \geq 0$, for all i?

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$$\lambda^* = \min_{\{i | u_i > 0\}} \frac{x_{B(i)}}{u_i}$$

$-\bar{c}_1$	$\dots$	$-\bar{c}_j$	$\dots$	$-\bar{c}_n$	$c_B^T x_B$
v_1	$\dots$	v_j	$\dots$	v_n	$x_{B(l)}$

$x_{B(l)} < 0$

$-\bar{c}_1 + v_1 \theta^*$	$\dots$	0	$\dots$	$-\bar{c}_n + v_n \theta^*$	$c_B^T x_B + x_{B(l)} \theta^*$
v_1 / v_j	$\dots$	1	$\dots$	v_n / v_j	$x_{B(l)} / v_j$

$$\theta^* = \min_{\{i | v_i < 0\}} \frac{\bar{c}_i}{v_i}.$$

*Dual Simple Method

Algorithm 5 Dual simplex method

Step 0. (Initialization) Find a basis matrix B for which the corresponding basic solution is dual feasible, i.e., $\bar{c}^T \leq 0$.

Step 1. (Optimality/primal feasibility check) Compute the primal solution $x_B = B^{-1}b$.
If $x_B \geq 0$, stop. The current solution is primal feasible and thus optimal.
Else, select some l such that $x_{B(l)} < 0$.

Step 2. (Minimum ratio test for dual feasibility) Consider the lth row of the tableau (the pivot row), which is of the form $(v_1, \ldots, v_n | x_{B(l)})$.
If $v_i \geq 0$ for all i, stop. The dual problem is unbounded and the primal problem is infeasible.
Else, compute the minimum ratio $\theta^* = \min_{\{i|v_i<0\}} \bar{c}_i / v_i$.
Let j be the column (the pivot column) that achieves the minimum ratio.
Then x_j enters the basis and $x_{B(l)}$ exits the basis.

Step 3. (Update the basis) Add to each row of the tableau a multiple of the lth row (the pivot row) so that v_j (the pivot element) becomes 1 and all other entries of the pivot column become 0.
Go to step 1.

Wyndor Glass Co. (Revisit)

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- Primal
  - x_j = # batches of product j produced per week, $j = 1, 2$
  - z = total profit per week

$$\begin{array}{lrrl} \max & z = & 3x_1 & +5x_2 \\ \text{s.t.} & & x_1 & \leq 4, \\ & & & 2x_2 \leq 12, \\ & & 3x_1 & +2x_2 \leq 18, \\ & & x_1, & x_2 \geq 0. \end{array}$$

- $x^* = (2, 6)$, $z^* = 36$

- What if…?

$$\begin{array}{lrrl} \max & z = & 6x_1 & +5x_2 \\ \text{s.t.} & & x_1 & \leq 4, \\ & & & 2x_2 \leq 12, \\ & & 3x_1 & +2x_2 \leq 18, \\ & & x_1, & x_2 \geq 0. \end{array}$$

$$\begin{array}{lrrl} \max & z = & 3x_1 & +5x_2 \\ \text{s.t.} & & x_1 & \leq 4, \\ & & & 2x_2 \leq 10, \\ & & 3x_1 & +2x_2 \leq 18, \\ & & x_1, & x_2 \geq 0. \end{array}$$

$$\begin{array}{lrrl} \max & z = & 3x_1 & +5x_2 \\ \text{s.t.} & & x_1 & \leq 4, \\ & & & 2x_2 \leq 12, \\ & & 3x_1 & +3x_2 \leq 18, \\ & & x_1, & x_2 \geq 0. \end{array}$$

Four Assumptions of LP (Revisit)

- Proportionality
  - Objective value/LHS is proportional to the level of activity
  - Nonlinear/mixed integer programming
- Additivity
  - Sum of individual contributions (objective/LHS)
  - Nonlinear programming
- Divisibility
  - (Mixed) Integer programming
- Certainty
  - Sensitivity analysis, but …
  - Stochastic programming, (approximate) dynamic programming, simulation optimization, robust optimization, …

Sensitivity Analysis

- A less ambitious approach, to study
  - dependence of the optimal value and the optimal solution on A, b, and c
  - conditions under which the current (optimal) basis remains optimal
  - how the optimal value changes (locally around the optimal solution)
- Changes in the right-hand side b
- Changes in the objective function c
- Changes in the consumption matrix A
- Adding a new variable
- Adding a new constraint

Primal-Dual Relationship (Revisit)

- Canonical form

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \geq c \\ & \pi \ u.r.s. \end{aligned}$$

- Optimal solution and value

$$\begin{aligned} x_B^* &= B^{-1} b \\ z* &= c_B^T B^{-1} b. \end{aligned}$$

- Optimal solution and value

$$\begin{aligned} (\pi^*)^T &= c_B^T B^{-1} \\ w^* &= c_B^T B^{-1} b. \end{aligned}$$

- x^* is optimal if

$$\text{Feasibility: } x_B = B^{-1} b \geq 0 \text{ (nonnegative solution).}$$

$$\text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).}$$

Changes in the Right-hand Side

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- Changing some component b from b_i to $b_i + \Delta$, or $b^{new} = b + \Delta e_i$
  - e_i is the ith unit vector

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- Changing some component b from b_i to $b_i + \Delta$, or $b^{new} = b + \Delta e_i$
  - e_i is the ith unit vector
  - Changing b does not affect the optimality condition.
  - The basis remains optimal if the feasibility condition is met.

$$\boxed{x_B^{new} = B^{-1}(b + \Delta e_i) = x_B + \Delta B^{-1} e_i \geq 0.}$$

  - Note that $B^{-1}e_i$ is the ith column of B^{-1}. Let $B^{-1} = [\beta_{ij}]$.

$$\boxed{\begin{array}{l} \text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).} \\ \\ \text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).} \end{array}}$$

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  - If violated (primal infeasible, but dual feasible), continue with the dual simplex method.

$$\boxed{\begin{array}{l} \text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).} \\ \\ \text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).} \end{array}}$$

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  - Note that $B^{-1}e_i$ is the ith column of B^{-1}. Let $B^{-1} = [\beta_{ij}]$.
  - The feasibility condition is

$$\boxed{\begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix} + \Delta \begin{bmatrix} \beta_{1i} \\ \vdots \\ \beta_{mi} \end{bmatrix} \geq 0.} \Rightarrow \boxed{\max_{\{j|\beta_{ji}>0\}} \left(-\frac{x_{B(j)}}{\beta_{ji}}\right) \leq \Delta \leq \min_{\{j|\beta_{ji}<0\}} \left(-\frac{x_{B(j)}}{\beta_{ji}}\right).}$$

  - If Δ is within the interval, the optimal value is

$$\boxed{z^{new,*} = c_B^T B^{-1}(b + \Delta e_i) = z^* + \pi_i^* \Delta.}$$

Changes in the Right-hand Side

- Example

$$
\begin{array}{llrrl}
\max & Z = & 3x_1 & +5x_2 & \\
\text{s.t.} & & x_1 & & \leq 4, \\
 & & & 2x_2 & \leq 12 \Rightarrow 10, \\
 & & 3x_1 & +2x_2 & \leq 18, \\
 & & x_1, & x_2 & \geq 0.
\end{array}
$$

Changes in the Right-hand Side

- Example

$$
\begin{array}{llrrl}
\max & Z = & 3x_1 & +5x_2 & \\
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 & & 3x_1 & +2x_2 & \leq 18, \\
 & & x_1, & x_2 & \geq 0.
\end{array}
$$

- The original problem in canonical form

$$
\begin{array}{llrrrrl}
\max & Z = & 3x_1 & +5x_2 & & & \\
\text{s.t.} & & x_1 & & +s_1 & & = 4, \\
 & & & 2x_2 & & +s_2 & = 12, \\
 & & 3x_1 & +2x_2 & & +s_3 & = 18, \\
 & & x_1, & x_2, & s_1, & s_2, \quad s_3 & \geq 0.
\end{array}
$$

Simplex Methods (Recall)

- Iteration 2
  - Step 3. Update solution and basis
    - Update the basis

$$c_B^T = \begin{bmatrix} 0 & 3 & 5 \end{bmatrix} \qquad c_N^T = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

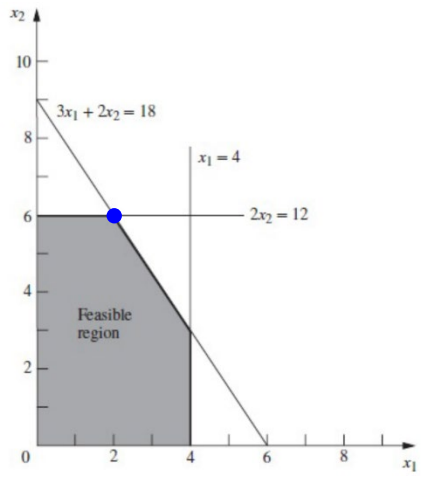
$$B = \begin{array}{c} \begin{array}{ccc} s_1 & x_2 & x_1 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 3 \end{bmatrix} \end{array} \qquad N = \begin{array}{c} \begin{array}{cc} s_3 & s_2 \end{array} \\ \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array} \qquad b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

    - Basic and nonbasic variables

$$B^{-1} = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix}$$

$$x_B = \begin{bmatrix} s_1 \\ x_2 \\ x_1 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \qquad x_N = \begin{bmatrix} s_3 \\ s_2 \end{bmatrix} = 0$$

$$\begin{array}{ll} \max & \begin{bmatrix} 3 & 5 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} \\ \text{s.t.} & \begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & s_1 & s_2 & s_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} \\ x \end{array} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \\ & \qquad\qquad\qquad\qquad\qquad\qquad \geq 0. \end{array}$$



Simplex Methods (Recall)

- Iteration 3
  - Step 1. Optimality check
    - Simplex multipliers

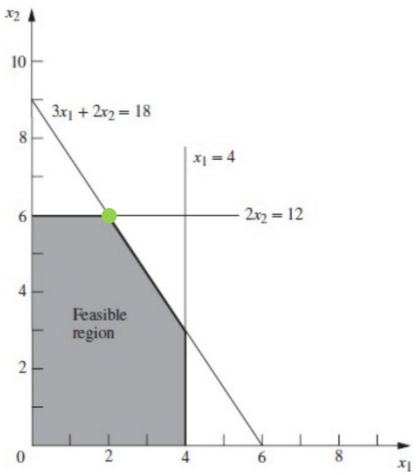
$$\pi^T = c_B^T B^{-1}$$
$$= \begin{bmatrix} 0 & 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 0 & 3/2 & 1 \end{bmatrix}$$

    - Reduced cost: improving directions?

$$\bar{c}_N^T = c_N^T - c_B^T B^{-1} N = c_N^T - \pi^T N$$
$$= \begin{bmatrix} 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 3/2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -3/2 \end{bmatrix}$$

    - No improving directions ➔ Optimal solution

$$\max \quad \begin{bmatrix} 3 & 5 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix}$$
$$\text{s.t.} \quad \begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & s_1 & s_2 & s_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \end{array} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$
$$x \geq 0.$$



Changes in the Right-hand Side

- Example

$$\begin{array}{llrrl} \max & Z = & 3x_1 & +5x_2 & \\ \text{s.t.} & & x_1 & & \leq 4, \\ & & & 2x_2 & \leq 12 \Rightarrow 10, \\ & & 3x_1 & +2x_2 & \leq 18, \\ & & x_1, & x_2 & \geq 0. \end{array}$$

- The original problem in canonical form

$$\begin{array}{llrrrrrl} \max & Z = & 3x_1 & +5x_2 & & & & \\ \text{s.t.} & & x_1 & & +s_1 & & & = 4, \\ & & & 2x_2 & & +s_2 & & = 12, \\ & & 3x_1 & +2x_2 & & & +s_3 & = 18, \\ & & x_1, & x_2, & s_1, & s_2, & s_3 & \geq 0. \end{array}$$

$$x_B = \begin{bmatrix} s_1 \\ x_2 \\ x_1 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \quad \text{and} \quad x_N = \begin{bmatrix} s_3 \\ s_2 \end{bmatrix} = 0$$

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 3 \end{bmatrix} \quad N = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix}$$

*Changes in the Right-hand Side

Example

$$\begin{array}{llrrl} \max & Z = & 3x_1 & +5x_2 & \\ \text{s.t.} & & x_1 & & \le 4, \\ & & & 2x_2 & \le 12 \Rightarrow 10, \\ & & 3x_1 & +2x_2 & \le 18, \\ & & x_1, & x_2 & \ge 0. \end{array}$$

- Changing b_2 ($\Delta = -2$), use the 2nd column of B^{-1}, the feasibility condition

$$\begin{array}{llrrrrrl} \max & Z = & 3x_1 & +5x_2 & & & & \\ \text{s.t.} & & x_1 & & +s_1 & & & = 4, \\ & & & 2x_2 & & +s_2 & & = 12, \\ & & 3x_1 & +2x_2 & & & +s_3 & = 18, \\ & & x_1, & x_2, & s_1, & s_2, & s_3 & \ge 0. \end{array}$$

$$x_B = \begin{bmatrix} s_1 \\ x_2 \\ x_1 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \quad \text{and} \quad x_N = \begin{bmatrix} s_3 \\ s_2 \end{bmatrix} = 0$$

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 3 \end{bmatrix} \qquad N = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad B^{-1} = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix}$$

$$\begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix} + \Delta \begin{bmatrix} \beta_{1i} \\ \vdots \\ \beta_{mi} \end{bmatrix} \ge 0. \quad \Rightarrow \quad \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} + \Delta \begin{bmatrix} 1/3 \\ 1/2 \\ -1/3 \end{bmatrix} \ge 0$$

*Changes in the Right-hand Side

- Example

$$\begin{array}{llrrl} \max & Z = & 3x_1 & +5x_2 & \\ \text{s.t.} & & x_1 & & \leq 4, \\ & & & 2x_2 & \leq 12 \Rightarrow 10, \\ & & 3x_1 & +2x_2 & \leq 18, \\ & & x_1, & x_2 & \geq 0. \end{array}$$

- Changing b_2 ($\Delta = -2$), use the 2nd column of B^{-1}, the feasibility condition

$$\begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} + \Delta \begin{bmatrix} 1/3 \\ 1/2 \\ -1/3 \end{bmatrix} \geq 0 \quad \Rightarrow \quad \Delta \in [-6, 6], \text{ or } b_2 \in [6, 18]$$

*Changes in the Right-hand Side

- Example

$$\begin{array}{llll} \max & Z = & 3x_1 & +5x_2 & \\ \text{s.t.} & & x_1 & & \leq 4, \\ & & & 2x_2 & \leq 12 \Rightarrow 10, \\ & & 3x_1 & +2x_2 & \leq 18, \\ & & x_1, & x_2 & \geq 0. \end{array}$$

- Changing b_2 ($\Delta = -2$), use the 2nd column of B^{-1}, the feasibility condition

$$\begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} + \Delta \begin{bmatrix} 1/3 \\ 1/2 \\ -1/3 \end{bmatrix} \geq 0 \quad \Rightarrow \quad \Delta \in [-6, 6], \text{ or } b_2 \in [6, 18]$$

- We have

$$x_B^{new} = \begin{bmatrix} s_1 \\ x_2 \\ x_1 \end{bmatrix} = B^{-1}(b + \Delta e_i) = x_B + \Delta B^{-1} e_i$$

$$= \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} + (-2) \begin{bmatrix} 1/3 \\ 1/2 \\ -1/3 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 5 \\ 8/3 \end{bmatrix}$$

$$\begin{aligned} z^{new,*} &= z^* + \pi_i^* \Delta \\ &= 36 + (3/2)(-2) = 33. \end{aligned}$$

Simplex Tableau (Recall)

x	s	RHS
$c_B^T B^{-1} A - c^T$	$c_B^T B^{-1}$	$c_B^T B^{-1} b$
$B^{-1} A$	B^{-1}	$B^{-1} b$

x	s	RHS
$-\bar{c}^T$	π^T	z
$B^{-1} A$	B^{-1}	x_B

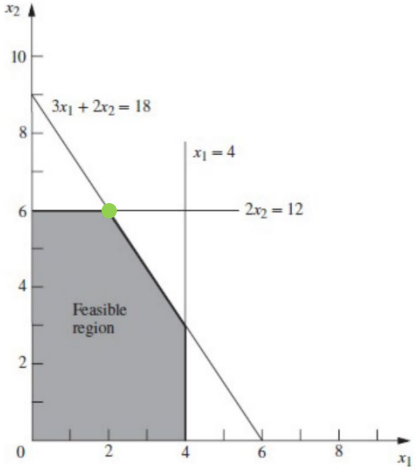
$$
\begin{array}{ll}
\max & \begin{bmatrix} 3 & 5 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} \\
\text{s.t.} & \\
& \begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & s_1 & s_2 & s_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} \\ x \end{array} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \\
& \qquad\qquad\qquad\qquad\qquad\qquad \geq 0.
\end{array}
$$

Iteration 3

- Step 1. Optimality check
  - Reduced cost

Basis	x1	x2	s1	s2	s3	RHS
Row 0	0	0	0	1 1/2	1	36
s1	0	0	1	1/3	- 1/3	2
x2	0	1	0	1/2	0	6
x1	1	0	0	- 1/3	1/3	2

  - No improving direction ➔ optimal solution



Changes in the Right-hand Side

Example

$$
\begin{array}{llrrl}
\max & Z = & 3x_1 & +5x_2 & \\
\text{s.t.} & & x_1 & & \leq 4, \\
& & & 2x_2 & \leq 12 \Rightarrow 10, \\
& & 3x_1 & +2x_2 & \leq 18, \\
& & x_1, & x_2 & \geq 0.
\end{array}
$$

- The original problem in canonical form

$$
\begin{array}{llrrrrrl}
\max & Z = & 3x_1 & +5x_2 & & & & \\
\text{s.t.} & & x_1 & & +s_1 & & & = 4, \\
& & & 2x_2 & & +s_2 & & = 12, \\
& & 3x_1 & +2x_2 & & & +s_3 & = 18, \\
& & x_1, & x_2, & s_1, & s_2, & s_3 & \geq 0.
\end{array}
$$

x	s	RHS
$-\bar{c}^T$	π^T	z
$B^{-1}A$	B^{-1}	x_B

Basis	x1	x2	s1	s2	s3	RHS
Row 0	0	0	0	1 1/2	1	36
s1	0	0	1	1/3	- 1/3	2
x2	0	1	0	1/2	0	6
x1	1	0	0	- 1/3	1/3	2

*Changes in the Right-hand Side

Example

$$
\begin{array}{llrrl}
\max & Z = & 3x_1 & +5x_2 & \\
\text{s.t.} & & x_1 & & \le 4, \\
& & & 2x_2 & \le 12 \Rightarrow 10, \\
& & 3x_1 & +2x_2 & \le 18, \\
& & x_1, & x_2 & \ge 0.
\end{array}
$$

- Changing b_2 ($\Delta = -2$), use the 2nd column of B^{-1}, the feasibility condition

$$
\begin{array}{llrrrrrl}
\max & Z = & 3x_1 & +5x_2 & & & & \\
\text{s.t.} & & x_1 & & +s_1 & & & = 4, \\
& & & 2x_2 & & +s_2 & & = 12, \\
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& & x_1, & x_2, & s_1, & s_2, & s_3 & \ge 0.
\end{array}
$$

x	s	RHS
$-\bar{c}^T$	π^T	z
$B^{-1}A$	B^{-1}	x_B

Basis	x1	x2	s1	s2	s3	RHS
Row 0	0	0	0	1 1/2	1	36
s1	0	0	1	1/3	- 1/3	2
x2	0	1	0	1/2	0	6
x1	1	0	0	- 1/3	1/3	2

$$\Delta \in [-6, 6], \text{ or } b_2 \in [6, 18]$$

$$
\begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix} + \Delta \begin{bmatrix} \beta_{1i} \\ \vdots \\ \beta_{mi} \end{bmatrix} \ge 0.
$$

$$
\max_{\{j \mid \beta_{ji} > 0\}} \left(-\frac{x_{B(j)}}{\beta_{ji}} \right) \le \Delta \le \min_{\{j \mid \beta_{ji} < 0\}} \left(\frac{x_{B(j)}}{\beta_{ji}} \right).
$$

Changes in the Objective Function

Changes in the Objective Function

- Changing some component of c from c_j to $c_j + \Delta$, or $c_j^{new} = c_j + \Delta$

*Changes in the Objective Function

- Changing some component of c from c_j to $c_j + \Delta$, or $c_j^{new} = c_j + \Delta$
  - Changing c does not affect the feasibility condition.
  - The basis remains optimal if the optimality condition holds.

$$\text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).}$$

$$\text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).}$$

Changes in the Objective Function

- Case 1: Changing the coefficient of a nonbasic variable x_j

$$\text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).}$$

$$\text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).}$$

*Changes in the Objective Function

- Case 1: Changing the coefficient of a nonbasic variable x_j
  - The only optimality condition that changes is the inequality for x_j.

$$\boxed{\bar{c}_j^{new} = (c_j + \Delta) - c_B^T B^{-1} A_j = \bar{c}_j + \Delta \leq 0.}$$

  - If violated (i.e., $\bar{c}_j^{new} > 0$), x_j becomes the entering variable.
  - Continue with the simplex method.

$$\boxed{\begin{array}{l} \text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).} \\ \\ \text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).} \end{array}}$$

Changes in the Objective Function

- Case 2: Changing the coefficient of a basic variable x_j

$$\text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).}$$

$$\text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).}$$

*Changes in the Objective Function

- Case 2: Changing the coefficient of a basic variable x_j
  - Let x_j be the kth basic variable in the optimal basis, that is, the basic variable associated with the kth constraint in the optimal basis.
  - Let $c_B^{new} = c_B + \Delta e_k$.
  - The optimality condition (for all nonbasic variables) is

$$\boxed{(\bar{c}_N^{new})^T = c_N^T - (c_B^T + \Delta e_k^T)B^{-1}N = \bar{c}_N^T - \Delta e_k^T B^{-1}N = \bar{c}_N^T - \Delta q_k \leq 0}$$

  - where $q_k = e_k^T B^{-1} N$ denotes the kth row of $B^{-1}N$.

$$\boxed{\begin{array}{l} \text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).} \\ \\ \text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).} \end{array}}$$

Changes in the Objective Function

- Example

$$
\begin{array}{llrrl}
\max & Z = & (3 \Rightarrow 6)x_1 & +5x_2 & \\
\text{s.t.} & & x_1 & & \leq 4, \\
 & & & 2x_2 & \leq 12, \\
 & & 3x_1 & +2x_2 & \leq 18, \\
 & & x_1, & x_2 & \geq 0.
\end{array}
$$

Changes in the Objective Function

- Example

$$
\begin{array}{lrrl}
\max & Z = & (3 \Rightarrow 6)x_1 \quad +5x_2 & \\
\text{s.t.} & & x_1 & \leq 4, \\
& & 2x_2 & \leq 12, \\
& & 3x_1 \quad +2x_2 & \leq 18, \\
& & x_1, \quad x_2 & \geq 0.
\end{array}
$$

  - The original problem in canonical form

$$
\begin{array}{lrl}
\max & Z = 3x_1 + 5x_2 & \\
\text{s.t.} & x_1 \quad +s_1 & = 4, \\
& 2x_2 \quad +s_2 & = 12, \\
& 3x_1 + 2x_2 \quad +s_3 & = 18, \\
& x_1, \quad x_2, \quad s_1, \quad s_2, \quad s_3 & \geq 0.
\end{array}
$$

Simplex Methods (Recall)

- Iteration 2
  - Step 3. Update solution and basis
    - Update the basis

$$c_B^T = \begin{bmatrix} 0 & 3 & 5 \end{bmatrix} \qquad c_N^T = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

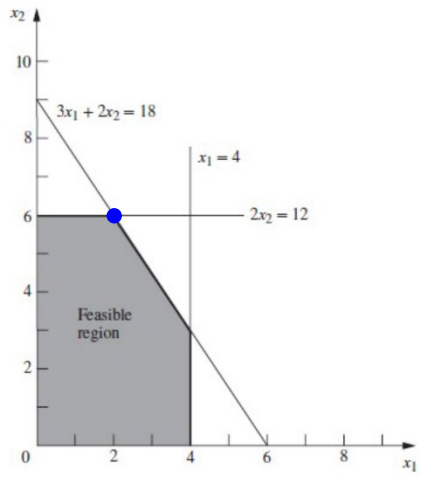
$$B = \begin{array}{c} \begin{array}{ccc} s_1 & x_2 & x_1 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 3 \end{bmatrix} \end{array} \qquad N = \begin{array}{c} \begin{array}{cc} s_3 & s_2 \end{array} \\ \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array} \qquad b = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

    - Basic and nonbasic variables

$$B^{-1} = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix}$$

$$x_B = \begin{bmatrix} s_1 \\ x_2 \\ x_1 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \qquad x_N = \begin{bmatrix} s_3 \\ s_2 \end{bmatrix} = 0$$

$$\begin{array}{ll} \max & \begin{bmatrix} 3 & 5 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} \\ \text{s.t.} & \begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & s_1 & s_2 & s_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} \\ x \end{array} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix} \\ & \qquad\qquad\qquad\qquad\qquad\qquad \geq 0. \end{array}$$



Simplex Methods (Recall)

- Iteration 3
  - Step 1. Optimality check
    - Simplex multipliers

$$\pi^T = c_B^T B^{-1}$$
$$= \begin{bmatrix} 0 & 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 0 & 3/2 & 1 \end{bmatrix}$$

    - Reduced cost: improving directions?

$$\bar{c}_N^T = c_N^T - c_B^T B^{-1} N = c_N^T - \pi^T N$$
$$= \begin{bmatrix} 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 3/2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -3/2 \end{bmatrix}$$

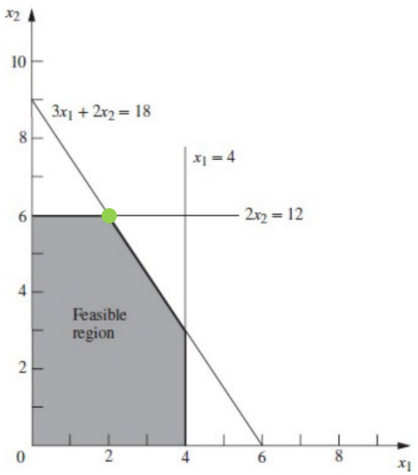
    - No improving directions ➔ Optimal solution

$$\max \quad \begin{bmatrix} 3 & 5 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix}$$

s.t.

$$\begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & s_1 & s_2 & s_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 1 \end{bmatrix} \end{array} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

$$x \geq 0.$$



Changes in the Objective Function

Example

$$\begin{array}{lrrl} \max & Z = & (3 \Rightarrow 6)x_1 + 5x_2 & \\ \text{s.t.} & & x_1 & \leq 4, \\ & & 2x_2 & \leq 12, \\ & & 3x_1 + 2x_2 & \leq 18, \\ & & x_1, \quad x_2 & \geq 0. \end{array}$$

- The original problem in canonical form

$$\begin{array}{lrrrrrl} \max & Z = & 3x_1 & +5x_2 & & & \\ \text{s.t.} & & x_1 & & +s_1 & & = 4, \\ & & & 2x_2 & & +s_2 & = 12, \\ & & 3x_1 & +2x_2 & & +s_3 & = 18, \\ & & x_1, & x_2, & s_1, & s_2, \quad s_3 & \geq 0. \end{array}$$

$$x_B = \begin{bmatrix} s_1 \\ x_2 \\ x_1 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \quad \text{and} \quad x_N = \begin{bmatrix} s_3 \\ s_2 \end{bmatrix} = 0$$

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 3 \end{bmatrix} \quad N = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix}$$

$$\bar{c}_N^T = \begin{bmatrix} -1 & -3/2 \end{bmatrix}$$

Changes in the Objective Function

Example

$$\begin{array}{lrrl} \max & Z = & (3 \Rightarrow 6)x_1 + 5x_2 & \\ \text{s.t.} & & x_1 & \le 4, \\ & & 2x_2 & \le 12, \\ & & 3x_1 + 2x_2 & \le 18, \\ & & x_1, \quad x_2 & \ge 0. \end{array}$$

- Changing c_1 ($\Delta = 3$), x_1 is the 3rd basic variable.

$$\begin{array}{lrl} \max & Z = 3x_1 + 5x_2 & \\ \text{s.t.} & x_1 + s_1 & = 4, \\ & 2x_2 + s_2 & = 12, \\ & 3x_1 + 2x_2 + s_3 & = 18, \\ & x_1, \quad x_2, \quad s_1, \quad s_2, \quad s_3 & \ge 0. \end{array}$$

$$x_B = \begin{bmatrix} s_1 \\ x_2 \\ x_1 \end{bmatrix} = B^{-1}b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \quad \text{and} \quad x_N = \begin{bmatrix} s_3 \\ s_2 \end{bmatrix} = 0$$

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*Changes in the Objective Function

- Example

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- Changing c_1 ($\Delta = 3$), x_1 is the 3rd basic variable.
- The optimality condition (for all nonbasic variables)

$$(\bar{c}_N^{new})^T = c_N^T - (c_B^T + \Delta e_k^T)B^{-1}N = \bar{c}_N^T - \Delta e_k^T B^{-1} N = \bar{c}_N^T - \Delta q_k \leq 0$$

- where $q_k = e_k^T B^{-1} N$ denotes the kth row of $B^{-1}N$.

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- Example

$$\begin{array}{lrrl} \max & Z = & (3 \Rightarrow 6)x_1 \quad +5x_2 & \\ \text{s.t.} & & x_1 & \leq 4, \\ & & 2x_2 & \leq 12, \\ & & 3x_1 \quad +2x_2 & \leq 18, \\ & & x_1, \quad x_2 & \geq 0. \end{array}$$

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- where $q_k = e_k^T B^{-1} N$ denotes the kth row of $B^{-1}N$.

$$B^{-1}N = \begin{bmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1/3 & 1/3 \\ 0 & 1/2 \\ \boxed{1/3 \quad -1/3} \end{bmatrix}$$

$$\bar{c}_N^T = \begin{bmatrix} -1 & -3/2 \end{bmatrix}$$

$$\bar{c}^{new} = \bar{c}_N^T - \Delta q_k = \begin{bmatrix} -1 \\ -3/2 \end{bmatrix} - \Delta \begin{bmatrix} 1/3 \\ -1/3 \end{bmatrix} \leq 0$$

$$\Delta \in [-3, 9/2], \text{ or } c_1 \in [0, 15/2]$$

$$z^{new,*} = z^* + \Delta x_j = 36 + (3)(2) = 42.$$

Changes in the Consumption Matrix

Changes in the Consumption Matrix

- Changing some entry in the jth column A_j of A from a_{ij} to $a_{ij} + \Delta$, or $A_j^{new} = A_j + \Delta e_i$

Changes in the Consumption Matrix

- Case 1: The column A_j is nonbasic.

*Changes in the Consumption Matrix

- Case 1: The column A_j is nonbasic.
  - The basis matrix B does not change, so the feasibility condition is unaffected.
  - Only the reduced cost of the jth column is affected.

$$\boxed{\bar{c}_j^{new} = c_j - c_B^T B^{-1}(A_j + \Delta e_i) = \bar{c}_j - \Delta c_B^T B^{-1} e_i = \bar{c}_j - \Delta \pi_i^T \leq 0}$$

  - If violated (i.e., $\bar{c}_j^{new} > 0$), x_j becomes the entering variable.
  - Continue with the simplex method.

$$\boxed{\begin{array}{l} \text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).} \\ \\ \text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).} \end{array}}$$

Changes in the Consumption Matrix

- Case 2: The column A_j is basic.

*Changes in the Consumption Matrix

- Case 2: The column A_j is basic.
  - Both feasibility and optimality conditions are affected.

$$\text{Feasibility: } x_B = B^{-1}b \geq 0 \text{ (nonnegative solution).}$$

$$\text{Optimality: } \bar{c}_N^T = c_N^T - c_B^T B^{-1} N \leq 0 \text{ (nonpositive reduced cost).}$$

Adding a New Variable

Adding a New Variable

- Canonical form

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

$$\Rightarrow$$

$$\begin{aligned} \max \quad & z = c^T x + c_{n+1} x_{n+1} \\ \text{s.t.} \quad & Ax + a_{n+1} x_{n+1} = b \\ & x, \quad x_{n+1} \geq 0. \end{aligned}$$

*Adding a New Variable

- Canonical form

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned} \quad \Rightarrow \quad \begin{aligned} \max \quad & z = c^T x + c_{n+1} x_{n+1} \\ \text{s.t.} \quad & Ax + a_{n+1} x_{n+1} = b \\ & x, \quad x_{n+1} \geq 0. \end{aligned}$$

  - $(x, x_{n+1}) = (x^*, 0)$ is a basic feasible solution to the new problem associated with the basis B, so we only need to examine the optimality condition, that is, the reduced cost of x_{n+1}.

$$\bar{c}_{n+1} = c_{n+1} - c_B^T B^{-1} a_{n+1} \leq 0.$$

  - If violated (i.e., $\bar{c}_{n+1} > 0$), x_{n+1} becomes the entering variable.
  - Continue with the simplex method.

Adding a New Constraint

Adding a New Constraint

- Adding a new constraint ➔ a restricted problem
  - If the current BFS satisfies the new constraint, then it is feasible to the new problem and thus optimal.
  - If the current BFS violates the new constraint (infeasible), introduce the constraint into the final simplex tableau (as an additional row)
    - Add a slack, surplus, or artificial variable, if needed.
    - Continue with the dual simplex method.

Q & A